

PROTOCOL: Polymerase Chain Reaction (PCR)

v 1.0

ONTOLOGY™ uses freeze-dried PCR plates that attaches unique molecular identifiers (UMIs) to the barcode region during PCR. The software recognizes these UMIs to automate demultiplexing. All components—except for water and DNA—are already in the plates. The user must just rehydrate, add DNA, and then thermocycle.

(1) Rehydrate ONTOLOGY™ PCR plates

- Prepare one freeze-dried PCR plate (orange) for each DNA plate (clear) to be sequenced. **Critical step: Ensure the plate numbers match—if you are using DNA plates #1-3 you must use PCR plates #1-3.**
- For each PCR plate (orange) to be used, use a reservoir and multichannel pipette to dispense **10 µL** of molecular-grade water to each well.
- Cover plate with a **temporary seal** and either quickly centrifuge or gently tap the plate against the bench to ensure all water moves to the bottom of the wells.
- Leave plate at room temperature for 10 min to allow PCR reagents to dissolve.
- Quickly centrifuge.

(2) Conduct PCR

- Slowly uncover the seal to avoid cross contamination**
- Transfer **2 µL** of neutralized lysate from the clear DNA plate into each well of the corresponding orange PCR plate; use a multichannel pipette and transfer row A → row A, row B → row B, etc.
Critical step: take extra care to ensure both plates are oriented the same way (A01 in top left corner), and that the number of the DNA plate and UMI plate match! Change tips between samples.
- Seal with a **PCR seal**
- Pulse down.
- Thermocycle using the ONTOLOGY™ PCR program:

	Temp (°C)	Duration	No. cycles
Denaturation/activation	94	2 min	1
Amplification 1	94	40 sec	5
	45	40 sec	
	72	1 min	
Amplification 2	94	40 sec	40
	51	40 sec	
	72	1 min	
Final elongation	72	2 min	1
Hold	4	hold	1



PCR plate components:

All PCR components except for the DNA are included in the freeze-dried plates; the user need only add water when ready to use. Each PCR contains the following reagents:

Component		1 rxn
Trehalose (10%)		6.25
Molecular grade water		2
Buffer (10X)		1.25
MgCl ₂ (50 mM)		0.625
Indexed Forward primer (10 uM) ¹		0.125
Indexed Reverse primer (10 uM) ²		0.125
dNTPs (10 mM)		0.0625
Platinum Taq (5U/uL)		0.06
DNA		2
TOTAL		12.5
Primer	Sequence (5' - 3')	
C_LepFolF ¹	RKTCAACMAATCATAAAGATATTGG	
C_LepFolR ²	TAAACTTCWGGRTGWCCAAAAAATC	

